

Differential Calculus — Chain Rule

Chapter 2: Exponential & Trigonometric Product | Page 33

PROBLEM STATEMENT

Product Rule & Multi-Chain Composite

Find the derivative with respect to x of the function:

$$y = e^{2x+1} \cdot \sin^3 x$$

STEP 1 প্রদত্ত গুণফলীয় ফাংশনটি উল্লেখ করি (Given Function)

$$y = e^{2x+1} \cdot \sin^3 x$$

STEP 2 উভয়পক্ষে x -এর সাপেক্ষে অন্তরীকরণ করি (Differentiating both sides w.r.t. x)

$$\frac{dy}{dx} = \frac{d}{dx} [e^{2x+1} \cdot \sin^3 x]$$

STEP 3 গুণন সূত্র বা Product Rule প্রয়োগ করি ($\frac{d}{dx}(uv) = uv' + vu'$)

$$\frac{dy}{dx} = e^{2x+1} \cdot \frac{d}{dx}(\sin^3 x) + \sin^3 x \cdot \frac{d}{dx}(e^{2x+1})$$

STEP 4 প্রতিটি অংশের ওপর পৃথকভাবে চেইন রুল প্রয়োগ করি (Evaluating Individual Derivatives)

$$\frac{d}{dx}(\sin^3 x) = 3 \sin^2 x \cdot \frac{d}{dx}(\sin x) = 3 \sin^2 x \cos x$$

$$\frac{d}{dx}(e^{2x+1}) = e^{2x+1} \cdot \frac{d}{dx}(2x+1) = 2e^{2x+1}$$

STEP 5 মান বসিয়ে সাধারণ উৎপাদক $e^{2x+1} \sin^2 x$ কমন নিই (Factoring Common Terms)

$$\frac{dy}{dx} = e^{2x+1} (3 \sin^2 x \cos x) + \sin^3 x (2e^{2x+1}) = e^{2x+1} \sin^2 x (3 \cos x + 2 \sin x)$$

FINAL ANSWER

$$\frac{dy}{dx} = e^{2x+1} \sin^2 x (3 \cos x + 2 \sin x)$$

📌 BOX 1: FORMULATION & PRODUCT-CHAIN INTERPLAY

- **Leibniz Product Formula:** $\frac{d}{dx}(u \cdot v) = u \frac{dv}{dx} + v \frac{du}{dx}$.
- For $u = e^{2x+1}$, $u' = 2e^{2x+1}$.
- For $v = \sin^3 x$, $v' = 3 \sin^2 x \cos x$.
- Factored form: $e^{2x+1} \sin^2 x (3 \cos x + 2 \sin x)$.

⚠️ BOX 2: COMMON STUDENT TRAPS & PITFALLS

- **Missing Factor of 2 in Exponential:** Differentiating e^{2x+1} to e^{2x+1} instead of $2e^{2x+1}$.
- **Stopping Before Factoring:** Leaving the un-factored sum $3e^{2x+1} \sin^2 x \cos x + 2e^{2x+1} \sin^3 x$ can cost presentation marks.

🌐 BOX 3: REAL DOMAIN & DIFFERENTIABILITY ANALYSIS

- **Domain of Definition:** Entire real line $\mathbb{R} = (-\infty, \infty)$.
- **Stationary Points:** $\frac{dy}{dx} = 0$ when:
 1. $\sin x = 0 \implies x = k\pi$ (multiplicity 2 roots).
 2. $3 \cos x + 2 \sin x = 0 \implies \tan x = -\frac{3}{2} \implies x = \arctan(-1.5) + k\pi$.

💡 BOX 4: LOGARITHMIC DIFFERENTIATION ALTERNATIVE

- Taking natural logarithm:

$$\ln y = \ln(e^{2x+1}) + \ln(\sin^3 x) = (2x+1) + 3 \ln(\sin x)$$

- Differentiating implicitly:

$$\frac{1}{y} \frac{dy}{dx} = 2 + 3 \left(\frac{\cos x}{\sin x} \right) = 2 + 3 \cot x = \frac{2 \sin x + 3 \cos x}{\sin x}$$

- Multiplying back by $y = e^{2x+1} \sin^3 x$:

$$\frac{dy}{dx} = e^{2x+1} \sin^3 x \left[\frac{3 \cos x + 2 \sin x}{\sin x} \right] = e^{2x+1} \sin^2 x (3 \cos x + 2 \sin x)$$

📊 BOX 5: AMIE EXAMINATION MARKS DISTRIBUTION (TOTAL: 4 / 5 MARKS)

- **1.0 Mark:** Correct statement of Leibniz product rule structure.
- **1.0 Mark:** Accurate exponential chain derivative $2e^{2x+1}$.
- **1.0 Mark:** Accurate trigonometric power derivative $3 \sin^2 x \cos x$.
- **1.0 Mark:** Factoring out $e^{2x+1} \sin^2 x$ and boxed final answer.